Grade 11 – Term 1
Choosing a Statistical Model: C10 Practice

1 Your Assignment

Write an **individual academic PDF article analyzing three different real-life situations**, each using a different statistical model studied in Term 1. Independently identify the situations and find **three credible, publicly accessible datasets containing actual observations**, one for each situation. Do not use invented or simulated datasets or copy completed practice problems.

Justify each model using the data and context, not simply a convenient formula. Apply methods studied in C1–C9 and show mathematical procedures even when using software.

2 Required Situations

1. **Linear probability density model.** Investigate a continuous variable that can reasonably be modeled by a linear probability density function over a stated interval. Specify its density, parameters, and support; verify nonnegativity and total area 1.
2. **Exponential model.** Investigate waiting times or intervals between events that can reasonably be modeled by an exponential distribution. Specify or estimate the rate parameter and justify a constant event rate and independent waiting times.
3. **Normal model and mean inference.** Investigate measurements that can reasonably be modeled by a normal distribution. Use the normal distribution or Student's t for inference about the population mean, as appropriate. State an inferential question and interpret your analysis in context. Justify the sampling assumptions and the plausibility of a normal population. Student's t is used for inference about a population mean when the population standard deviation is unknown and estimated from the sample, with $\nu = n - 1$ degrees of freedom. Under appropriate assumptions, it describes the standardized sample-mean statistic; it is **not the distribution of the original observations merely because the sample is small**.

3 Requirements for Each Situation

Complete all eight requirements for **each of the three situations**:

1. **Describe the context.** Explain the real-life situation, define the random variable and its units, identify the population and relevant time period, and state the objective of the analysis.
2. **Find real data.** Provide a working, clickable hyperlink to the original database or dataset. Identify the source, sample size, relevant variables, and how observations were collected or selected. Explain any filtering or missing-value treatment. Cite the source where used.
3. **Organize the data.** Present the observations or an appropriate frequency table with clear headings and units. For a large dataset, use a frequency table and explain how it was constructed.
4. **Create a histogram.** Construct a readable histogram from the dataset with meaningful class intervals, frequencies, a descriptive title, and labeled axes and units. Explain your interval choices. If overlaying a theoretical curve, use compatible scales: a density histogram or a curve rescaled to frequencies.
5. **Select and justify the model.** Specify the distribution, support, and parameters, distinguishing known values, estimates, and assumptions. Justify the choice using the context, data characteristics, and histogram. Discuss

assumptions and limitations. A histogram provides empirical evidence but **does not, by itself, establish that a theoretical distribution is appropriate**. Explain any important mismatch and qualify your conclusions.

6. **Calculate sample statistics.** Calculate the sample mean, sample variance, and sample standard deviation using

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}.$$

   Show substitutions and intermediate steps; software results alone are insufficient. State units, including squared units for variance. Prefer original observations for calculations; if using grouped data, show the frequency-based procedure and label midpoint-based results as approximations.

7. **Calculate theoretical statistics.** Whenever parameters can reasonably be specified or estimated, determine the model's theoretical mean, variance, and standard deviation using the formulas or methods studied. Show parameters, formulas, substitutions, and procedures. Clearly distinguish model values from sample statistics and label values based on estimated parameters as estimates. For Situation 3, compare observations with the underlying normal measurement model; keep Student's t inference separate. If a parameter cannot reasonably be determined, explain the missing information and how it limits the comparison.

8. **Compare and interpret.** Compare the histogram and sample statistics with the theoretical model. Explain agreements and differences, considering sampling variation, parameter estimation, and assumptions that may fail. If fitting makes a model statistic equal to a sample statistic, acknowledge that this equality is not independent evidence of fit. Discuss whether the evidence supports the model and interpret the findings and limitations in context.

4 Required Article Structure

Write a connected academic article using the following structure. Integrate data, graphs, and calculations into the discussion rather than submitting disconnected answers.

Title and Author Give a descriptive title and your full name.

Introduction Explain the article's purpose and introduce the three situations and their objectives.

Methodology Describe the datasets, sources, variables, sample sizes, data preparation, model selection, assumptions, parameter estimates, and statistical procedures.

Results and Discussion Use three clearly identified subsections: **Situation 1: Linear Density**, **Situation 2: Exponential**, and **Situation 3: Normal and Mean Inference**. Each must contain the data or frequency table, histogram, calculations, justification, comparisons, interpretations, and limitations required above.

Conclusion Summarize the findings and compare the usefulness and limitations of the three models.

References Give complete citations and clickable hyperlinks to all three original datasets: author or organization, dataset title, date or version where available, and access date. Cite any other sources and tools used, and connect in-text citations to the reference list.

5 Before You Submit

Submit one **readable, professionally organized PDF article written individually**. Check that all three situations use the required different approaches and real datasets, model choices are justified, and mathematical procedures are shown. Use clear headings, complete paragraphs, and consistent notation. Open the final PDF to verify that all graphs, tables, and equations are complete and readable. Test all three original dataset hyperlinks to confirm that they work.